

Stylometry Based AI Authorship Detection: Methodology and Mathematical Foundations

Minor Project Report Document · Text Input · Decision Tree, Random Forest, SVM, Naive Bayes and a Proposed Hybrid

1. Purpose and Scope

This document sets out the methodology and the mathematics behind the project. The central argument is that the classical classifiers (Decision Tree, Random Forest, SVM, Naive Bayes) usually fail in ordinary AI detection setups for identifiable mathematical reasons, that stylometric features address several of those reasons, and that a carefully designed hybrid model can combine the strengths of lexical and stylistic evidence.

The Streamlit application is not the contribution. It is an explanatory instrument: each page is built to make one mathematical idea visible (a split criterion, a margin, a likelihood contribution, a calibration curve). The research contribution is the experimental and theoretical analysis.

One caution applies to the whole document. The claims below are hypotheses with a mathematical rationale, not established results. Every claim is paired with an experiment in Section 10 that can confirm or refute it, and negative results are reported as findings.

2. Problem Formulation

Let a sample be x (a text) with true label y in $\{0, 1\}$, where 0 means human authored and 1 means AI generated. A feature map ϕ converts x into a vector $\phi(x)$ in $\mathbb{R}^d$. A classifier learns $f: \mathbb{R}^d \rightarrow [0, 1]$ or a score $s(x)$, and predicts 1 when the score passes a threshold t .

The quantity we care about is the expected risk under the deployment distribution, not the training distribution:

$$\begin{aligned} R_{\text{deploy}}(f) &= E \text{ over } (x, y) \sim P_{\text{deploy}} [\text{loss}(f(\phi(x))), y] \\ R_{\text{train}}(f) &= E \text{ over } (x, y) \sim P_{\text{train}} [\text{loss}(f(\phi(x))), y] \end{aligned}$$

AI detection fails mainly because P_{deploy} differs from P_{train} (new topics, new generators, edited text, short inputs) while the classifier was optimised for P_{train} . This is distribution shift, and it is the lens used throughout the document.

It helps to decompose a sample into two parts:

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x = (c, s)
c = content: topic, vocabulary tied to the subject, named entities
s = style: how the content is expressed (rhythm, punctuation, structure,
function words)
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A good detector should depend on s and be approximately invariant to c. A detector that depends on c learns the dataset, not the author type. Most failures discussed below are failures to be invariant to c.

3. Why Detectors Fail in Practice: Four Root Causes

Root cause	Mathematical description	Effect
Shortcut learning	Some feature z is correlated with y in training ($P_{\text{train}}(y \text{ given } z)$ is high) but not causally linked to authorship	High test score in domain, collapse out of domain
Covariate shift	$P_{\text{train}}(x)$ differs from $P_{\text{deploy}}(x)$ while $P(y \text{ given } x)$ is assumed fixed	Learned thresholds and vocabulary weights no longer fit
Generator drift	The AI class is a moving target: $P(x \text{ given } y = 1)$ changes with every new model version	A detector fitted to one generator misses another
High dimension with small sample	d is much larger than n for TF IDF features	Overfitting, unstable weights, inflated confidence

The corpus HC3 illustrates the first cause. Human and ChatGPT answers differ in length, formatting and collection process, so length alone can separate them. A classifier can score highly by learning length, which says nothing about authorship style. The shortcut diagnostic in Section 10 measures exactly this.

4. Classifier by Classifier: Mathematics and Failure Modes

4.1 Decision Tree

How it works. A tree recursively partitions the feature space with axis aligned splits of the form $x_j \leq \tau$. At each node it chooses the feature j and threshold τ that maximise impurity reduction. For a node with class proportions p_0 and p_1 :

$$\begin{aligned} \text{Gini}(\text{node}) &= 1 - (p_0^2 + p_1^2) \\ \text{Entropy}(\text{node}) &= -(p_0 \log_2 p_0 + p_1 \log_2 p_1) \\ \text{Gain}(j, \tau) &= \text{Impurity}(\text{parent}) - (n_L/n \cdot \text{Impurity}(\text{left}) + n_R/n \cdot \text{Impurity}(\text{right})) \end{aligned}$$

The split with the largest gain is chosen greedily and the process repeats until a stopping rule (maximum depth, minimum leaf size) is met.

Why it fails in AI detection.

1. High variance. A fully grown tree can reach zero training error. By the bias variance decomposition, expected error = bias² + variance + noise, and an unpruned tree has near zero bias but very high variance: a small change in the training data changes the chosen root split and therefore the whole tree.
2. Greedy shortcut selection. The first split is the single feature with the largest gain. On a corpus where length separates the classes, the root split is a length threshold. Everything beneath it is then fitted to the residual, so the tree is built around the shortcut.
3. Sparse TF IDF is a poor fit. On a matrix with tens of thousands of mostly zero columns, the best split is often a rare token present in a handful of documents. Such a split has real gain on training data but is a topic or source artefact.
4. Axis aligned boundaries. Style is a joint property (for example, long sentences together with few commas together with high lexical repetition). A single tree needs many splits to approximate a diagonal boundary, which inflates depth and variance.
5. Hard step functions. A leaf outputs a constant, so predicted probabilities are coarse and poorly calibrated.

How stylometry helps. Stylometric features are dense, continuous and low dimensional (tens, not tens of thousands). Thresholds such as "mean sentence length ≤ 14.2 " are meaningful and stable, so splits are less likely to be rare token artefacts. Limiting depth

and leaf size then controls variance in a principled way. The tree also becomes interpretable, which suits the explanatory goal of the application.

4.2 Random Forest

How it works. A forest averages B trees, each trained on a bootstrap sample with a random subset of m features considered at each split (typically $m = \sqrt{d}$). The prediction is the average vote:

$$f_{\text{RF}}(x) = (1/B) \cdot \sum_b T_b(x)$$

If each tree has variance σ^2 and the pairwise correlation between trees is ρ , the variance of the average is:

$$\text{Var}(f_{\text{RF}}) = \rho \cdot \sigma^2 + (1 - \rho) \cdot \sigma^2 / B$$

As B grows the second term vanishes, but the first term remains. The forest only reduces variance to the extent that trees are decorrelated.

Why it fails in AI detection.

1. Persistent correlation from a dominant feature. If length, a formatting marker or a topic word is strongly predictive, most trees use it, so ρ stays high and the variance floor $\rho \cdot \sigma^2$ does not shrink. Averaging many trees that make the same shortcut mistake does not remove the mistake. It also lowers variance without lowering bias: if the shortcut is wrong on new data, the forest is confidently wrong.
2. No extrapolation. Trees predict piecewise constants learned from observed ranges. For an input outside the training range (a much shorter text, an unfamiliar generator) the forest returns the value of the nearest leaf, with no signal that it is out of range.
3. Poor probability estimates. Averaged votes are biased away from 0 and 1 and are not calibrated probabilities, so a displayed score of 0.9 does not mean 90 percent correct.
4. Sparse high dimensional input weakens the random feature subset. With d very large and most features uninformative, a random subset of m features often contains no useful feature, which lowers the strength of individual trees.

How stylometry helps. With a compact set of meaningful features, the random subset at each split is far more likely to contain informative features, so individual trees are stronger. Feature importance (mean decrease in impurity, or permutation importance)

becomes readable: it ranks stylistic families rather than anonymous vocabulary columns. Permutation importance should be preferred because impurity importance is biased toward high cardinality continuous features.

4.3 Support Vector Machine

How it works. A linear SVM finds the hyperplane $w \cdot x + b = 0$ that separates classes with the largest margin. With labels coded as y in $\{-1, +1\}$, the soft margin objective is:

$$\min \text{ over } w, b : \quad (1/2) \cdot ||w||^2 + C \cdot \sum_i \max(0, 1 - y_i (w \cdot x_i + b))$$

The margin width is $2/||w||$. The first term maximises the margin, the second (hinge loss) penalises points inside the margin or misclassified, and C trades the two off. Classical generalisation bounds scale with R^2/γ^2 , where R is the radius of the data and γ the margin, so the bound depends on margin and scale rather than directly on the number of features. In the dual form, the solution depends only on inner products, which is what allows kernels $K(x, x') = \phi(x) \cdot \phi(x')$.

Why it fails in AI detection.

1. Weights bind to vocabulary. On TF IDF features, each coordinate of w is a word or character n gram weight. The hyperplane is therefore a weighted list of tokens. When the topic or generator changes, the active tokens change, and the learned w no longer points in a useful direction.
2. Scores are not probabilities. The decision value $f(x) = w \cdot x + b$ is a signed distance, not $P(y = 1 \text{ given } x)$. Platt scaling fits $P(y = 1 \text{ given } x) = 1 / (1 + \exp(A \cdot f(x) + B))$, but A and B are estimated on one distribution and are only as reliable as that calibration set is representative.
3. Sensitivity to feature scale. The margin is computed in Euclidean distance, so unscaled numeric features (character count in the hundreds, ratios between 0 and 1) let large scale features dominate. Standardisation is mandatory for stylometric inputs.
4. Near perfect separability in high dimension. When $d \gg n$, a linear separator almost always exists, so training accuracy is near 100 percent whether or not the features carry real authorship signal. A high training score therefore says little.
5. Length coupling. TF IDF is L2 normalised per document, but short and long texts still produce differently shaped vectors, so margin behaviour changes with input length.

How stylometry helps. Scaled stylometric features give a low dimensional, content independent input space where the weight vector has a direct reading: w_j says how much mean sentence length, comma rate or lexical diversity pushes toward the AI class.

The margin geometry can be drawn in two dimensions in the application. An RBF kernel can capture mild nonlinearity, but for explanation a linear kernel is preferred.

4.4 Naive Bayes

How it works. Naive Bayes applies Bayes' rule with a conditional independence assumption:

$$P(y \text{ given } x) \propto P(y) \cdot \prod_j P(x_j \text{ given } y)$$

In the multinomial form for word counts, with θ_{yj} the probability of term j in class y and Laplace smoothing α :

$$\theta_{yj} = (N_{yj} + \alpha) / (N_y + \alpha \cdot V)$$

$$\log P(y \text{ given } x) = \log P(y) + \sum_j x_j \cdot \log \theta_{yj} + \text{constant}$$

The class score is a sum of per term log likelihood contributions, which is why the model is easy to explain term by term. For continuous features, Gaussian Naive Bayes uses $P(x_j \text{ given } y) = N(\mu_{yj}, \sigma^2_{yj})$.

Why it fails in AI detection.

1. The independence assumption is false for language. Words co occur strongly, so the log likelihoods being summed are correlated. The same evidence is counted several times, which pushes posterior probabilities toward 0 or 1. Naive Bayes can classify reasonably but its probabilities are badly overconfident.
2. Length dependence. The log likelihood sum grows with the number of tokens, so longer documents yield more extreme scores. Naive Bayes scores are not comparable across lengths, which hurts mixed length detection.
3. Vocabulary mismatch under shift. A term unseen for one class gets probability mass only from smoothing. On a new topic or generator, many terms are rare or absent, and the result is dominated by the prior and by smoothing choices.
4. Gaussian assumption on stylometry. Features such as sentence length or type token ratio are skewed or bounded, so a normal density is a poor model, and many stylometric features are strongly correlated with each other, violating independence again.
5. Discriminative information is discarded. Because it models $P(x \text{ given } y)$ rather than the boundary, capacity is spent on describing each class, not separating them.

How stylometry helps. Naive Bayes is most defensible on a small set of weakly correlated stylometric features after transformation (log or Box Cox) toward normality, and after removing redundant features using a correlation threshold. It is retained mainly as an honest baseline: its failure patterns are mathematically clear and useful for the report.

4.5 Summary Table

Classifier	Optimises	Main weakness on raw TF IDF	Effect of stylometry
Decision Tree	Greedy impurity gain	High variance, rare token and length shortcuts	Meaningful dense thresholds, interpretable
Random Forest	Variance reduction by averaging	Correlated trees, no extrapolation, poor calibration	Stronger trees, readable importance
SVM	Margin plus hinge loss	Weights tied to vocabulary, scale sensitivity, uncalibrated	Scaled content independent geometry
Naive Bayes	Class conditional likelihood	Independence violation, overconfidence, length effect	Better on transformed, decorrelated features

5. How Stylometry Addresses These Failures

5.1 The Core Argument

Stylometric features are statistics of how text is written rather than what it is about. Under the decomposition $x = (c, s)$, they are designed so that $\phi_{\text{style}}(x)$ depends mainly on s . If that holds, then $P(\phi_{\text{style}}(x) \text{ given } y)$ changes less across topics than $P(\phi_{\text{tfidf}}(x) \text{ given } y)$ does, which directly reduces the covariate shift term in the generalisation gap. Three further mathematical benefits follow.

1. Lower dimension, lower variance. A model with $d = 50$ features trained on n samples has a much smaller capacity than one with $d = 50,000$, so the variance term of the bias variance tradeoff falls.

2. Features have an interpretation, so explanations are about measurable properties and can be checked.
3. Features can be normalised for length by construction, which removes the length shortcut when done carefully.

5.2 Text Stylometric Features with Definitions

Let N be the number of word tokens, V the number of distinct words, V_1 the number of words occurring once (hapax legomena), and V_k the number of word types occurring exactly k times.

Type token ratio adjust it)	$TTR = V / N$	(falls with N , so length
Root TTR (Guiraud)	$R = V / \sqrt{N}$	
Hapax ratio	$H = V_1 / V$	
Yule's K length stable)	$K = 10^4 \cdot (\sum_k k^2 \cdot V_k - N) / N^2$	(largely
Honore's statistic	$R_H = 100 \cdot \log N / (1 - V_1 / V)$	
Sentence lengths $L_1..L_m$:		
mean, median, standard deviation, range		
Burstiness $B = (\sigma_L - \mu_L) / (\sigma_L + \mu_L)$		
Punctuation rate for mark q :	$r_q = \text{count}(q) / (\text{number of characters})$	
Function word frequency:	$f_w = \text{count}(w) / N$ for w in a fixed list	
Character entropy: distribution	$H_c = - \sum_a p_a \log_2 p_a$ over character	

Burstiness is of particular interest: human writing tends to mix short and long sentences, while generated text is often more uniform. This is a hypothesis to test, not an assumption, and it can reverse for edited or prompted outputs.

Practical rules: use rates and ratios rather than raw counts, preserve punctuation and capitalisation in the stylometry branch, and fix the function word list in advance so that train and inference use the same vocabulary.

5.3 Function Word Isolation (FW RoBERTa and FW PLL)

This family scores style on the words that carry the least topic information. A part of speech tagger (for example spaCy) splits tokens into content words (nouns, verbs, adjectives) and function words (prepositions, conjunctions, auxiliaries, pronouns,

determiners). Generative models tend to pick predictable, safe grammatical connectors, a tendency called neural text degeneration, so function word choice may differ between human and AI text even when the topic is the same.

Two feature routes are proposed. Let F be the set of function word positions in a text of T tokens.

FW RoBERTa: run a pretrained RoBERTa over the whole text, keep only the hidden states h_t at positions t in F ,
and pool them into one document vector
$$h_F = (1/|F|) \cdot \sum_{t \in F} h_t$$

FW PLL: pseudo log likelihood restricted to function words, using the masked language model
$$PLL_F(x) = (1/|F|) \cdot \sum_{t \in F} \log P_{MLM}(w_t \text{ given all other tokens })$$

The working hypothesis is that AI text has a higher mean PLL_F (more predictable connectors) and a smaller spread across sentences. This is tested, not assumed.

Three cautions apply. First, the content words are excluded from the pooled vector, but the contextual hidden states still see them, so some topic information can leak in; the leave one topic out experiment in Section 10 is what shows whether the features really are domain invariant. Second, the notes this section draws on report about 86.6 percent accuracy on unseen topics and new LLM versions; that figure is a reported claim to reproduce, not a result of this project. Third, a pretrained transformer is used only as a frozen feature extractor, so the classifiers stay classical (a linear SVM on h_F), but the pipeline is no longer purely handcrafted. Keep T2 as the handcrafted comparator so the effect of this addition can be measured.

5.4 Trigram Stylometry

Three trigram families capture habits at different levels. Each is stored as relative frequencies, so text length does not leak into the features:

$f_g = \text{count}(\text{trigram } g) / \text{total trigrams in the text}$ (the vector sums to 1)

1. Character trigrams: every run of three characters, including spaces and punctuation. They capture sub word patterns and punctuation habits without depending on language boundaries.

2. Syntactic trigrams: parse each sentence into a dependency tree (spaCy or StanfordNLP), traverse it breadth first, and extract parent child triples of tags or relations. They capture how clauses are structured rather than which words are used.
3. Word trigrams: three word sequences, including the transition phrases that LLMs are reported to overuse (the notes mention words such as significant, notable and despite, and phrases such as recent years).

Two practical points. The trigram vocabulary (the top K most frequent trigrams) is fitted on training data only, and K is tuned on validation data. Because the vector is compositional, apply a log or square root transform before a linear model.

Phrase lists age quickly. A list of overused phrases learned from one generation of models is a form of generator drift risk, so word trigrams are the family most likely to fail on a newer generator, and the held out generator test in Section 10 is how to find out.

5.5 Statistical Redundancy, Perplexity and Burstiness

Bayesian n gram redundancy. For n from 1 to 4, measure how much of the text is repeated n grams. Let r_n be the share of n gram tokens that occur more than once (its complement is the singularity, the share that occur exactly once). Compare the two class models with a Bayes factor and add the evidence in log space:

$$\begin{aligned} BF_n &= P(r_n \text{ given AI}) / P(r_n \text{ given human}) \\ \log \text{ posterior odds} &= \log \text{ prior odds} + \sum_{n=1..4} \log BF_n \end{aligned}$$

The sum treats the four orders as independent, but they overlap heavily, so it double counts evidence and is overconfident in the same way as Naive Bayes (Section 4.4). Use the log odds as one input to the meta learner, never as a final probability. The notes report that this unsupervised redundancy signal separates human and LLM text in English and French and across academic and creative domains; that is a claim to reproduce here.

Perplexity and burstiness. With a reference language model, sentence perplexity is:

$$\begin{aligned} PPL(\text{sentence}) &= \exp(-(1/T) \cdot \sum_t \log P(w_t \text{ given } w_{\text{before}_t})) \\ B &= (\sigma_p - \mu_p) / (\sigma_p + \mu_p) \quad \text{over the sentence perplexities } p_1..p_m \end{aligned}$$

The mean captures overall predictability, B captures how uneven it is, and the variance of the gaps between repeated tokens (token inter arrival variance) captures how clumped repetition is. The hypothesis is that AI text is uniformly predictable: low mean, low variance, B near its low end.

The caution is serious. Perplexity depends on the reference model and on text type, and formal or heavily edited human writing also has low perplexity. Perplexity based detectors are a documented source of false positives against non native English writers, so this family must be checked for subgroup false positive rates, not only overall accuracy.

5.6 Topic Evolution Graph (TEM 2)

Stylometry captures how a text is written. This family captures how ideas flow from one part of the text to the next. Split the document into paragraphs P_1 to P_m , represent each as the mean of its Word2Vec word vectors e_i , and build a directed graph whose nodes are paragraphs. An edge from i to j (with i before j) exists when the cosine similarity passes a threshold τ :

$$\text{sim}(i, j) = (e_i \cdot e_j) / (||e_i|| \cdot ||e_j||) \quad \text{edge } i \rightarrow j \text{ if } \text{sim}(i, j) \geq \tau$$

The hypothesis is that human writers branch, add tangents and return to earlier themes, while LLM text tends to be a flat, tightly connected single thread. The graph is turned into connectivity features (the notes call them TEGMs):

1. Edge density: edges divided by the possible edges.
2. Mean out degree and the number of branching nodes (out degree of 2 or more).
3. Longest chain ratio: length of the longest path divided by m .
4. Mean shortest path length, number of connected components and clustering coefficient.

Three limits matter. The graph needs enough paragraphs (a minimum such as 4, set on validation data), so short answers have undefined metrics and must go through the length gate in Section 7. The threshold τ and the embedding choice are hyperparameters tuned on validation data only. And HC3 answers are mostly short, so this family may need a longer form corpus to be tested fairly.

5.7 Burrows's Delta as a Reference Method

The repository [jamesosullivan/stylometry](https://github.com/jamesosullivan/stylometry) collects Python scripts for Burrows's Delta (with MDS and dendrogram plots) and for SentenceTransformer document embeddings with PCA. It was built to attribute literary texts to known human authors, not to detect AI, so it supplies methods to adapt rather than a ready detector. It is MIT licensed and asks to be cited by author, institution and software title, so cite it if its code or design is reused.

The mathematics. Let k be the number of most frequent words (MFW), chosen on training data only. Let f_{iw} be the occurrences of word w in text i per 1,000 tokens, and let μ_w and σ_w be the mean and standard deviation of f_{iw} over the training texts.

$$z_{iw} = (f_{iw} - \mu_w) / \sigma_w$$

$$\Delta(i, j) = (1/k) \cdot \sum_{w \text{ in MFW}} |z_{iw} - z_{jw}| \quad (\text{Manhattan distance on } z \text{ scores})$$

Z scoring gives every word equal weight, so very common words such as "the" do not dominate. The MFW are mostly function words, so Δ is a function word method and connects directly to Section 5.3. The repository normalises to frequency per 1,000 tokens by default to reduce text length effects, the same reason this project uses rates rather than counts.

Adapting it to detection.

1. Nearest centroid classifier. Take the mean z vector of each class in training (c_{human} and c_{AI}) and score a new text by $\Delta(x, c_{\text{human}}) - \Delta(x, c_{\text{AI}})$. With Euclidean distance the boundary is a hyperplane (a special case of linear discriminant analysis with identity covariance). With the Manhattan distance of Δ it is piecewise linear and less sensitive to one extreme word frequency. Calibrate the score like the other base learners and add it to the stacked hybrid as a further base learner.
2. AI is not one author. Each generator behaves like a separate author, so a single AI centroid blurs them. Build one centroid per generator, report results per generator, and test on a held out generator.
3. Distance variant. A cosine version of Δ (angle between z vectors) is a common alternative that is often more stable when text lengths vary. Treat the choice of distance and of k (for example 100 to 500) as hyperparameters tuned on validation data.

Visual tools for the application. MDS places each text as a point in two dimensions so that plotted distances approximate the Δ distances d_{ij} , by minimising $\text{stress} = \sqrt{\sum (d_{ij} - \hat{d}_{ij})^2 / \sum d_{ij}^2}$. The dendrogram shows hierarchical clustering of the same distance matrix. Both belong on the Stylometry Explorer page: if human and AI texts separate on the MDS plot, the audience sees the geometry behind the nearest centroid rule.

Embeddings, and why they are a useful warning. The repository states that SentenceTransformer embeddings capture semantic and stylistic signals together, so PCA clusters are not pure style. That is the content plus style mixing of Section 2 seen in

practice. Plain document embeddings are therefore a good deliberate baseline: if they cluster by topic but not by class, they show why FW RoBERTa masks content words. On limited hardware, the repository suggests a smaller MiniLM model in place of the large RoBERTa default, and it splits long texts into chunks of a set number of words, which links to the length rules in Section 7.

Implementation advice. Re implement the Delta function inside the project's own feature module with unit tests, rather than depending on the scripts. They read one folder of text files per author and auto install packages, whereas this project uses JSONL records with group metadata. Fit the MFW list, mu and sigma on the training split only, or the test set leaks into the features.

5.8 An Honest Limit

Stylometry does not guarantee robustness. Generators change, prompts can request a casual voice, and light human editing moves a sample toward human statistics. Stylometric features reduce certain shortcut and shift problems but are themselves vulnerable to style imitation and to short inputs, where estimates such as sigma_L are noisy (the standard error of a sample mean falls as $1/\sqrt{m}$). This is a main reason the project includes reliability and abstention logic.

6. The Hybrid Model

6.1 Why a Hybrid Is Needed

Lexical features (TF IDF, token n grams) and stylometric features carry partly different information. Lexical features capture generator specific phrasing and common templates (high in domain accuracy but fragile). Stylometric features capture structure and rhythm (more portable but weaker in isolation and noisy on short inputs). Combining them is worthwhile only if their errors are not perfectly correlated, which is something the experiments must measure.

6.2 Proposed Design: Reliability Aware Stacked Hybrid (RASH)

The proposal has two layers, with an optional early fusion variant for comparison.

Variant A: Early fusion (feature level). Concatenate blocks with a weight per block:

$$\text{phi_H}(x) = [\text{alpha} \cdot \text{phi_tfidf}(x) , \text{beta} \cdot \text{phi_style_scaled}(x)]$$

For an SVM this is equivalent to a sum of kernels, which is itself a valid kernel because a nonnegative combination of positive semidefinite kernels is positive semidefinite:

$$K_H(x, x') = \alpha^2 \cdot K_{tfidf}(x, x') + \beta^2 \cdot K_{style}(x, x')$$

The weights α and β are chosen on the validation set. Without block weighting, the many TF IDF columns dominate the few stylometric columns in both the margin and in the L2 penalty, which silently turns the hybrid back into a lexical model. That is why block scaling is part of the design and not an optional detail.

Variant B: Late fusion by stacking (decision level), the proposed main model.

1. Base learners (level 0): a linear SVM on FW RoBERTa vectors; a Random Forest or LightGBM on explicit stylometric metrics (type token ratio, sentence length variance, punctuation ratios, topic graph metrics); XGBoost or logistic regression on normalised character, syntactic and word trigram distributions; a Bayesian classifier on the n gram redundancy log odds; and a Burrows's Delta nearest centroid learner (Section 5.7). A linear SVM on TF IDF is kept as the lexical reference learner, so the ablation shows what the new learners add.
2. Each base learner produces an out of fold score on the training data (using grouped k fold, so related samples never straddle folds). Scores are calibrated, for example with Platt scaling, and converted to log odds $z_m = \log(p_m / (1 - p_m))$.
3. A meta learner (L2 regularised logistic regression, or XGBoost as an alternative) combines them:

$$P(y = 1 \text{ given } x) = \text{sigmoid}(w_0 + \sum_m w_m \cdot z_m(x))$$

4. A reliability layer uses the spread of the z_m values (disagreement), input length against the validated minimum and a feature range check to decide between a qualified prediction and abstention.

Out of fold training of the meta learner is essential. If base learners score the same data they were fitted on, their scores are optimistic and the meta learner learns to trust them too much (leakage).

6.3 Why the Hybrid Can Beat Stylometry Alone

1. Complementary errors. For an ensemble of M predictors with average individual error and average pairwise error correlation ρ , variance of the combined score is approximately $\sigma^2 \cdot (\rho + (1 - \rho)/M)$. The gain is largest when ρ is small, which is

plausible when one learner uses lexical evidence and another uses style. The experiments report the measured correlation of errors, so the claim is tested, not assumed.

2. Higher in domain accuracy. Lexical evidence recovers generator specific signals that coarse style statistics miss, so the hybrid should dominate stylometry alone on in domain data.
3. Robustness to short inputs. When the text is too short for stable style statistics, the meta learner can place more weight on features that remain informative. A gating variable based on length can let w_m depend on input length.
4. Disagreement as a reliability signal. Divergence between a lexical learner and a stylistic learner is useful evidence of distribution shift: if they disagree, the input is likely unlike the training data. Stylometry alone has no such internal cross check.
5. Smoother bias variance tradeoff. Stylometry alone is low variance but can underfit (high bias). Lexical alone is low bias in domain but high variance. The hybrid occupies a middle position, tuned by alpha and beta or by the meta weights.

6.4 Why the Hybrid Can Also Fail (to be reported honestly)

1. If the lexical block still carries a shortcut, the hybrid may learn it and lose the cross domain benefit of stylometry. The block ablation in Section 10 checks this.
2. Agreement among learners trained on the same biased data is not evidence of correctness.
3. More components mean more hyperparameters and a higher risk of overfitting the validation set. Nested or repeated grouped cross validation controls this.
4. If cross domain results show the hybrid equal to or worse than stylometry alone, that is a legitimate result, and the recommended deployment becomes stylometry plus abstention.

6.5 Comparison Summary

Property	Baseline (TF IDF)	Stylometry only	Hybrid (RASH)
Dimension	Very high	Low	Mixed, block weighted
In domain accuracy	High	Moderate	Highest expected

Property	Baseline (TF IDF)	Stylometry only	Hybrid (RASH)
Topic and generator shift	Fragile	More robust (to be tested)	Depends on block weights, tested by ablation
Short inputs	Moderate	Noisy	Gated, with abstention
Interpretability	Token weights	Named features	Per component plus meta weights
Reliability signal	Score only	Score only	Score plus disagreement

7. Reliability: Calibration and Abstention

A score is not a probability unless calibrated. Calibration is measured by the expected calibration error over B bins:

$$\begin{aligned} \text{ECE} &= \sum_b \left(n_b / n \right) \cdot | \text{acc}(b) - \text{conf}(b) | \\ \text{Brier} &= (1/n) \cdot \sum_i \left(p_i - y_i \right)^2 \\ \text{LogLoss} &= -(1/n) \cdot \sum_i \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right] \end{aligned}$$

Selective prediction lets the system abstain on uncertain inputs. For a confidence function $\kappa(x)$ and threshold t :

$$\begin{aligned} \text{Coverage}(t) &= \text{fraction of inputs with } \kappa(x) \geq t \\ \text{SelectiveRisk}(t) &= \text{error rate among inputs with } \kappa(x) \geq t \end{aligned}$$

Plotting selective risk against coverage gives the risk coverage curve. A useful system shows risk falling as coverage is reduced. A system that abstains on everything difficult achieves low risk at near zero coverage and is not useful, so report both. Thresholds are chosen on validation data only, then frozen.

SVM scores need Platt or isotonic calibration. Naive Bayes and Random Forest probabilities typically need calibration too. All calibration is fitted on data not used to fit the classifier.

7.1 Length Gating

Stylometric metrics such as sentence length variance and type token ratio are noisy on short inputs, because the standard error of a sample mean shrinks only as $1/\sqrt{m}$. The notes suggest a starting rule: below about 150 words or 10 sentences, return "Unsure,

insufficient sample" instead of guessing. Treat those numbers as starting values. Choose the minimum length on validation data by plotting accuracy and ECE against length bins and taking the smallest length at which error stays under a target. Features with their own minimums (the topic graph needs a minimum number of paragraphs) get their own gates.

7.2 Model Disagreement

When the lexical and stylistic learners predict opposite classes, treat it as a sign of distribution shift. Measure disagreement as the spread of the calibrated log odds, $D(x)$ = standard deviation of $z_m(x)$ over the base learners, and abstain when $D(x)$ passes a threshold chosen on validation data. Agreement is not proof of correctness, since learners trained on the same biased data can agree and still be wrong.

7.3 Post Editing Quality (xCOMET)

Users may paraphrase or translate text to evade detection. xCOMET is a neural metric that scores the semantic quality of the edited text against the original on a scale from 0 to 1. The notes report that stylometric detectors keep about 95 percent accuracy while xCOMET stays above 0.55, and fall below 80 percent when xCOMET drops under 0.30, where the text is close to unreadable. These are claims to reproduce, not results of this project.

The test is to paraphrase and round trip translate the test texts at several strengths, score each against its original with xCOMET, bin by score, and plot detection accuracy per bin with confidence intervals. Two limits: xCOMET needs the original as a reference, which exists in an experiment but not for a user submission, so it is a tool for robustness analysis rather than a runtime feature; and it adds a neural model dependency.

8. Experimental Methodology

8.1 Research Questions

ID	Question	Method
RQ1	Do stylometric features improve text detection over TF IDF?	Compare T1, T2, T3
RQ2	Which classifier performs best on identical splits?	DT, RF, SVM, NB on shared folds

ID	Question	Method
RQ3	Do gains persist on unseen topics, sources or generators?	Cross domain tests
RQ4	Can the system recognise unreliable predictions?	Calibration, abstention, risk coverage
RQ5	Does the hybrid improve generalisation over stylometry alone?	RASH against T2, with paired tests

8.2 Data and Splits

1. Text: HC3 as a starting corpus, with a second source or generator held out for external testing.
2. Common schema: sample_id, content, label, task_id, source_id, generator, language, edited. Metadata fields never enter the classifier.
3. Group aware splitting by task so answers to the same prompt stay together (GroupKFold or GroupShuffleSplit). Remove exact duplicates and investigate near duplicates.
4. Three partitions: training, validation (hyperparameters, calibration, thresholds) and an untouched test set used once.

8.3 Three Levels of Evaluation

1. In domain: train and test on the same distribution with clean splits.
2. Cross domain: leave one topic, task, source or generator out. Report separately because each measures a different kind of generalisation.
3. Robustness: paraphrased, edited, reformatted, mixed origin and translated samples (semantic quality scored with xCOMET), with each transformation and its severity documented.

8.4 Metrics

Accuracy, precision and recall for the AI class, macro F1, ROC AUC, PR AUC, false positive rate (humans flagged as AI), Brier score, log loss, ECE, coverage and selective risk, with confusion matrices and per domain breakdowns. The false positive rate deserves emphasis because wrongly accusing a human author is the costliest error.

8.5 Statistical Treatment

- 1. Repeated grouped cross validation, reporting mean and standard deviation across folds.
- 2. Bootstrap confidence intervals on test metrics by resampling groups (tasks), not individual rows.
- 3. Paired comparison on the same test samples. For two classifiers, McNemar’s test uses the discordant counts b (A correct, B wrong) and c (A wrong, B correct):

$$\chi^2 = (|b - c| - 1)^2 / (b + c) \qquad \text{(one degree of freedom, with continuity correction)}$$

- 4. Correct for multiple comparisons when many models are tested (for example Holm correction).

8.6 Ablations

Remove one family at a time (length, punctuation, lexical diversity, function words, character n grams, function word embeddings, trigram families, n gram redundancy, topic graph metrics) and for the hybrid vary alpha and beta from 0 to 1. This shows which block carries the signal and whether the hybrid is secretly a lexical model.

9. The Application as an Explanatory Tool

Each page of the Streamlit app is designed to teach one mathematical idea, using the trained models as live examples.

Page	What it shows	Mathematical idea illustrated
Authorship Analyzer	Paste text, see a qualified result, warnings and top features	End to end pipeline, input validation, abstention
Classifier Lab	Four tabs, one per classifier	See below
Stylometry Explorer	Feature distributions for human and AI classes, with overlap, plus an MDS plot and dendrogram of Burrows’s Delta distances	Why features separate classes, and where they overlap

Page	What it shows	Mathematical idea illustrated
Hybrid Lab	Sliders for alpha and beta, stacking weights, error correlation matrix	Kernel sum, complementary errors, ensemble variance
Reliability Lab	Calibration curve, risk coverage curve, short input and shift tests	ECE, selective risk, distribution shift
Shortcut Diagnostic	Accuracy of a length only or metadata only model	Shortcut learning
Model Comparison	Tables, confusion matrices, paired test results	Statistical comparison under identical splits
Paragraph Attribution	Highlights paragraphs where authorship appears to switch in mixed submissions	Segment level scoring, and why short segments give noisy scores
Feature Highlights	Colour coded function word density, overused transition phrases and sentence length distribution	Style versus content, burstiness
Topic Evolution Graph	Directed graph of how themes develop across paragraphs	Graph connectivity metrics used as features

The Classifier Lab tabs:

1. Decision Tree: draw the tree, display Gini and gain at each split, and let the user change maximum depth to watch variance rise (train and validation gap).
2. Random Forest: show how the number of trees and the correlation between trees change variance, and display permutation importance.
3. SVM: show a two dimensional PCA projection with the margin, and a slider for C to show margin width against training error.
4. Naive Bayes: show for one input the per term log likelihood contribution ($\log \theta_{1j} - \log \theta_{0j}$), and the effect of correlated terms on the posterior.

Good practice: models are trained offline and loaded as versioned artifacts, submitted content is not stored by default, and every export records model version, dataset version and limitations.

10. Experiments That Test the Claims in This Document

Claim	Experiment	Evidence of support
Classifiers exploit shortcuts	Train on length only, source only and topic only features	A metadata only model scoring far above chance indicates a shortcut
TF IDF models fail under topic shift	Hold out entire topics (leave one domain out)	Large drop from in domain to cross domain scores
Random Forest averaging does not remove a shared shortcut	Measure tree correlation and score with and without length features	Little change in variance floor, large change in score without length
SVM weights track vocabulary	Inspect top weights, test on a new generator	Top weights are topic words, performance falls
Naive Bayes is overconfident	Calibration curve and ECE for each classifier	Highest ECE and a posterior piled at 0 and 1
Stylometry transfers better	Compare T1 and T2 across held out domains	Smaller in domain to cross domain gap for T2
Hybrid beats stylometry alone	T3 against T2 in domain and cross domain, paired tests, error correlation	Significant gain, with error correlation below 1
Abstention helps	Risk coverage curve on test	Selective risk falls as coverage falls, beyond a random baseline
Function word features carry style, not topic	Leave one topic out comparison of FW RoBERTa and FW PLL features against TF IDF	Smaller drop from in domain to cross domain than TF IDF
Topic flow separates human from AI text	Topic graph metrics alone on long form text, then added to the hybrid	Above chance on long text, and a gain in the ablation
Short inputs are unreliable	Accuracy and ECE by length bin	Error falls with length, giving a justified minimum length

Claim	Experiment	Evidence of support
Detection holds under paraphrase until quality collapses	Paraphrase and translation at several strengths, accuracy against xCOMET bin	Stable accuracy at high quality, falling at low quality
Delta captures function word style	Burrows's Delta nearest centroid, k from 100 to 500, per generator centroids, leave one topic and one generator out	Above chance, stable across topics, with the generator gap reported

Any row that does not support its claim is reported as a negative result with an explanation.

11. Report Structure

1. Introduction and motivation, with the problem of false accusations.
2. Related work in stylometry, AI text detection and calibration.
3. Problem formulation and the shift view (Sections 2 and 3 of this document).
4. Classifier mathematics and failure analysis (Section 4).
5. Stylometric feature design (Section 5).
6. Hybrid model design (Section 6).
7. Datasets, splits and leakage controls.
8. Experimental protocol, metrics and statistics (Section 8).
9. Results, ablations and failure analysis (Section 10).
10. The explanatory application (Section 9).
11. Limitations, ethics and future work.

In the limitations, state clearly that detectors can produce false positives, that results hold only for the studied generators, domains and languages, and that no output of this system should be treated as proof of authorship.

12. Suggested Timeline

Week	Focus
1	Research questions, datasets, schema, deduplication, grouped splits, shortcut diagnostic
2	TF IDF baselines, four classifiers, fixed seeds, saved predictions
3	Text stylometry features, unit tests, stylometry only classifiers
4	Hybrid early and late fusion, tuning, ablations, cross domain tests
5	Calibration, abstention, risk coverage, Streamlit pages that explain the mathematics
6	Frozen test evaluation, failure analysis, final report and demo

13. Summary of the Argument

Decision Trees overfit and chase shortcuts, Random Forests average away variance but not a shared bias, SVMs tie their weights to vocabulary and produce uncalibrated scores, and Naive Bayes counts correlated evidence repeatedly and becomes overconfident. All four are hurt by distribution shift when fed high dimensional lexical features. Stylometric features reduce dimension, remove much of the dependence on content and give interpretable inputs, but they are noisy on short inputs and are still vulnerable to generator drift. A calibrated, stacked hybrid that combines lexical and stylistic evidence, and abstains when its components disagree or the input is out of range, is the proposed way to gain accuracy while being honest about uncertainty. Whether it actually does so on held out data is the question the experiments are designed to answer.